OSPF Features and Characteristics

1.1.1

Introduction to OSPF

This topic is a brief overview of Open Shortest Path First (OSPF), which includes single-area and multiarea. OSPFv2 is used for IPv4 networks. OSPFv3 is used for IPv6 networks. The primary focus of this entire module is single-area OSPFv2.

OSPF is a link-state routing protocol that was developed as an alternative for the distance vector Routing Information Protocol (RIP). RIP was an acceptable routing protocol in the early days of networking and the Internet. However, the RIP reliance on hop count as the only metric for determining best route quickly became problematic. Using hop count does not scale well in larger networks with multiple paths of varying speeds. OSPF has significant advantages over RIP in that it offers faster convergence and scales to much larger network implementations.

OSPF is a link-state routing protocol that uses the concept of areas. A network administrator can divide the routing domain into distinct areas that help control routing update traffic. A link is an interface on a router. A link is also a network segment that connects two routers, or a stub network such as an Ethernet LAN that is connected to a single router. Information about the state of a link is known as a link-state. All link-state information includes the network prefix, prefix length, and cost.

This module covers basic, single-area OSPF implementations and configurations.

1.1.2

Components of OSPF

All routing protocols share similar components. They all use routing protocol messages to exchange route information. The messages help build data structures, which are then processed using a routing algorithm.

Click each OSPF component below for more information.

Routing Protocol Messages

Routers running OSPF exchange messages to convey routing information using five types of packets. These packets, as shown in the figure, are as follows:

- Hello packet
- Database description packet
- Link-state request packet
- Link-state update packet
- Link-state acknowledgment packet

These packets are used to discover neighboring routers and also to exchange routing information to maintain accurate information about the network.

Data Structures

Algorithm

1.1.3

Link-State Operation

To maintain routing information, OSPF routers complete a generic link-state routing process to reach a state of convergence. The figure shows a five router topology. Each link between routers is labeled with a cost value. In OSPF, cost is used to determine the best path to the destination. The following are the link-state routing steps that are completed by a router:

1. Establish Neighbor Adjacencies
2. Exchange Link-State Advertisements
3. Build the Link State Database
4. Execute the SPF Algorithm
5. Choose the Best Route

Click each button for an illustration of the steps in the link-state routing process that R1 uses to reach convergence.

1. Establish Neighbor Adjacencies

2. Exchange Link-State Advertisements

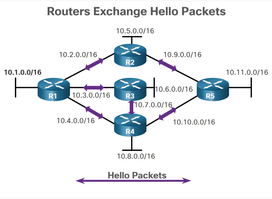
3. Build the Link State Database

4. Execute the SPF Algorithm

5. Choose the Best Route

1. Establish Neighbor Adjacencies

OSPF-enabled routers must recognize each other on the network before they can share information. An OSPF-enabled router sends Hello packets out all OSPF-enabled interfaces to determine if neighbors are present on those links. If a neighbor is present, the OSPF-enabled router attempts to establish a neighbor adjacency with that neighbor.



1.1.4

Single-Area and Multiarea OSPF

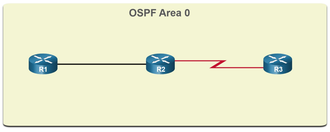
To make OSPF more efficient and scalable, OSPF supports hierarchical routing using areas. An OSPF area is a group of routers that share the same link-state information in their LSDBs. OSPF can be implemented in one of two ways, as follows:

- **Single-Area OSPF** – All routers are in one area. Best practice is to use area 0.
- **Multiarea OSPF** – OSPF is implemented using multiple areas, in a hierarchical fashion. All areas must connect to the backbone area (area 0). Routers interconnecting the areas are referred to as Area Border Routers (ABRs).

The focus of this module is on single-area OSPFv2.

Click each button to compare single-area and multiarea OSPF.

Single-Area OSPF | **Multiarea OSPF**



1.1.5

Multiarea OSPF

With multiarea OSPF, one large routing domain can be divided into smaller areas, to support hierarchical routing. Routing still occurs between the areas (interarea routing), while many of the processor intensive routing operations, such as recalculating the database, are kept within an area.

For instance, any time a router receives new information about a topology change within the area (including the addition, deletion, or modification of a link) the router must rerun the SPF algorithm, create a new SPF tree, and update the routing table. The SPF algorithm is CPU-intensive and the time it takes for calculation depends on the size of the area.

Note: Routers in other areas receive updates regarding topology changes, but these routers only update the routing table, not rerun the SPF algorithm.

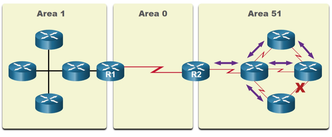
Too many routers in one area would make the LSDBs very large and increase the load on the CPU. Therefore, arranging routers into areas effectively partitions a potentially large database into smaller and more manageable databases.

The hierarchical-topology design options with multiarea OSPF can offer the following advantages.

- **Smaller routing tables** – Tables are smaller because there are fewer routing table entries. This is because network addresses can be summarized between areas. Route summarization is not enabled by default.
- **Reduced link-state update overhead** – Designing multiarea OSPF with smaller areas minimizes processing and memory requirements.
- **Reduced frequency of SPF calculations** – Multiarea OSPF localize the impact of a topology change within an area. For instance, it minimizes routing update impact because LSA flooding stops at the area boundary.

For example, in the figure R2 is an ABR for area 51. A topology change in area 51 would cause all area 51 routers to rerun the SPF algorithm, create a new SPF tree and update their IP routing tables. The ABR, R2, would send an LSA to routers in the area 0, which would eventually be flooded to all routers in the OSPF routing domain. This type of LSA does not cause routers in other areas to rerun the SPF algorithm. They only have to update their LSDB and routing table.

Link Change Impacts Local Area Only



- Link failure affects the local area only (area 51).
- The ABR (R2) isolates the flooding of a specific LSA to area 51.
- Routers in areas 0 and 1 do not need to run the SPF algorithm.

1.1.6

OSPFv3

OSPFv3 is the OSPFv2 equivalent for exchanging IPv6 prefixes. Recall that in IPv6, the network address is referred to as the prefix and the subnet mask is called the prefix-length.

Similar to its IPv4 counterpart, OSPFv3 exchanges routing information to populate the IPv6 routing table with remote prefixes.

Note: With the OSPFv3 Address Families feature, OSPFv3 includes support for both IPv4 and IPv6. OSPF Address Families is beyond the scope of this curriculum.

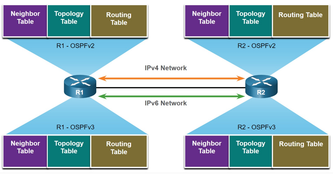
OSPFv2 runs over the IPv4 network layer, communicating with other OSPF IPv4 peers, and advertising only IPv4 routes.

OSPFv3 has the same functionality as OSPFv2, but uses IPv6 as the network layer transport, communicating with OSPFv3 peers and advertising IPv6 routes. OSPFv3 also uses the SPF algorithm as the computation engine to determine the best paths throughout the routing domain.

OSPFv3 has separate processes from its IPv4 counterpart. The processes and operations are basically the same as in the IPv4 routing protocol, but run independently. OSPFv2 and OSPFv3 each have separate adjacency tables, OSPF topology tables, and IP routing tables, as shown in the figure.

The OSPFv3 configuration and verification commands are similar to those used in OSPFv2.

OSPFv2 and OSPFv3 Data Structures



1.1.7

Check Your Understanding – OSPF Features and Characteristics

Check your understanding of OSPF features and characteristics by choosing the BEST answer to the following questions.

1. Which of the following OSPF components is associated with the neighbor table?

- Dijkstra's algorithm
- Link-State database
- Routing protocol messages
- (•) Adjacency database
- Forwarding database

2. Which of the following OSPF components is responsible for computing the cost of each route?

- (•) Dijkstra's algorithm
- Link-State database
- Routing protocol messages
- Adjacency database
- Forwarding database

3. Which of the following OSPF components is associated with the topology table?

- Dijkstra's algorithm
- (•) Link-State database
- Routing protocol messages
- Adjacency database
- Forwarding database

4. Which of the following OSPF components is associated with the routing table?

- Dijkstra's algorithm
- Link-State database
- Routing protocol messages
- Adjacency database
- (•) Forwarding database

5. What is the correct order in the steps for Link-State operation?

- 1. Execute the SPF Algorithm
  2. Establish Neighbor Adjacencies
  3. Exchange Link-State Advertisements
  4. Choose the Best Route
  5. Build the Topology Table
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